

T2 Minería de datos

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Carga de librerías

```
packages <- c(
  "tidyverse",
  "tidymodels",
  "dplyr",
  "glmnet",
  "MASS",
  "broom",
  "pROC",
  "yardstick",
  "palmerpenguins",
  "titanic",
  "modeldata",
  "factoextra",
  "cluster",
  "ggrepel"
)
#to_install <- packages[!sapply(packages, requireNamespace, quietly = TRUE)]
#
#if (length(to_install) > 0) {
#  install.packages(to_install, repos = "https://cloud.r-project.org")
#}
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.3      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
library(tidymodels)

## -- Attaching packages ----- tidymodels 1.5.0 --
## v broom      1.0.13      v rsample     1.3.2
## v dials      1.4.4      v tailor      0.1.0
## v infer      1.1.0      v tune        2.1.0
## v modeldata  1.5.1      v workflows   1.3.0
## v parsnip    1.6.0      v workflowsets 1.1.1
```

```

## v recipes      1.3.3      v yardstick    1.4.0
## -- Conflicts ----- tidymodels_conflicts() --
## x scales::discard() masks purrr::discard()
## x dplyr::filter()   masks stats::filter()
## x recipes::fixed() masks stringr::fixed()
## x dplyr::lag()      masks stats::lag()
## x yardstick::spec() masks readr::spec()
## x recipes::step()   masks stats::step()

library(dplyr)
library(glmnet)

## Cargando paquete requerido: Matrix
##
## Adjuntando el paquete: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Loaded glmnet 5.0

library(MASS)

##
## Adjuntando el paquete: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select

library(broom)
library(pROC)

## Type 'citation("pROC")' for a citation.
##
## Adjuntando el paquete: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var

library(yardstick)
library(palmerpenguins)

##
## Adjuntando el paquete: 'palmerpenguins'
##
## The following object is masked from 'package:modeldata':
##
##     penguins
##
## The following objects are masked from 'package:datasets':
##
##     penguins, penguins_raw

```

```
library(titanic)
library(modeldata)
library(factoextra)
```

```
## Welcome to factoextra!
```

```
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practice
```

```
library(cluster)
library(ggrepel)
```

1.1. Exploración y limpieza de datos a) Cargue el dataset y describa sus dimensiones, tipos de variables y porcentaje de valores perdidos por columna. Viendo datos, dimensiones y estructura

```
data("titanic_train")
```

```
head(titanic_train)
```

```
##   PassengerId Survived Pclass
## 1           1         0       3
## 2           2         1       1
## 3           3         1       3
## 4           4         1       1
## 5           5         0       3
## 6           6         0       3
```

```
##                                     Name    Sex Age SibSp Parch
## 1                                Braund, Mr. Owen Harris  male  22     1     0
## 2 Cumings, Mrs. John Bradley (Florence Briggs Thayer) female  38     1     0
## 3                                Heikkinen, Miss. Laina female  26     0     0
## 4 Futrelle, Mrs. Jacques Heath (Lily May Peel) female  35     1     0
## 5                                Allen, Mr. William Henry  male  35     0     0
## 6                                Moran, Mr. James       male  NA     0     0
```

```
##           Ticket    Fare Cabin Embarked
## 1      A/5 21171  7.2500      S
## 2      PC 17599 71.2833    C85      C
## 3 STON/O2. 3101282  7.9250      S
## 4      113803 53.1000   C123      S
## 5      373450  8.0500      S
## 6      330877  8.4583      Q
```

```
dim(titanic_train)
```

```
## [1] 891 12
```

```
str(titanic_train)
```

```
## 'data.frame':   891 obs. of  12 variables:
##  $ PassengerId: int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Survived   : int  0 1 1 1 0 0 0 0 1 1 ...
##  $ Pclass     : int  3 1 3 1 3 3 1 3 3 2 ...
##  $ Name       : chr  "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)"
##  $ Sex        : chr  "male" "female" "female" "female" ...
##  $ Age        : num  22 38 26 35 35 NA 54 2 27 14 ...
##  $ SibSp      : int  1 1 0 1 0 0 0 3 0 1 ...
##  $ Parch      : int  0 0 0 0 0 0 0 1 2 0 ...
##  $ Ticket     : chr  "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
##  $ Fare       : num  7.25 71.28 7.92 53.1 8.05 ...
##  $ Cabin      : chr  "" "C85" "" "C123" ...
```

```
## $ Embarked : chr "S" "C" "S" "S" ...
```

```
titanic_train$Embarked
```

```
## [1] "S" "C" "S" "S" "S" "Q" "S" "S" "S" "C" "S" "S" "S" "S" "S" "S" "Q" "S"
## [19] "S" "C" "S" "S" "Q" "S" "S" "S" "C" "S" "Q" "S" "C" "C" "Q" "S" "C" "S"
## [37] "C" "S" "S" "C" "S" "S" "C" "C" "Q" "S" "Q" "Q" "C" "S" "S" "S" "C" "S"
## [55] "C" "S" "S" "C" "S" "S" "C" "S" "S" "S" "C" "C" "S" "S" "S" "S" "S" "S"
## [73] "S" "C" "S" "S" "S" "S" "S" "S" "S" "S" "Q" "S" "S" "S" "S" "S" "S"
## [91] "S" "S" "S" "S" "S" "S" "C" "C" "S" "S" "S" "S" "S" "S" "S" "S" "S"
## [109] "S" "Q" "S" "C" "S" "S" "C" "S" "Q" "S" "C" "S" "S" "S" "C" "S" "S" "C"
## [127] "Q" "S" "C" "S" "C" "S" "S" "S" "S" "C" "S" "S" "S" "C" "C" "S" "S" "Q"
## [145] "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "C" "Q" "S" "S" "S" "S" "S"
## [163] "S" "S" "S" "S" "S" "S" "S" "S" "S" "Q" "S" "S" "C" "S" "S" "C" "S" "S"
## [181] "S" "C" "S" "S" "S" "S" "Q" "S" "Q" "S" "S" "S" "S" "S" "C" "C" "Q" "S"
## [199] "Q" "S" "S" "S" "S" "C" "S" "S" "S" "C" "Q" "C" "S" "S" "S" "S" "Q" "C"
## [217] "S" "S" "C" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S"
## [235] "S" "S" "S" "S" "S" "S" "C" "Q" "S" "S" "C" "Q" "S" "S" "S" "S" "S" "S"
## [253] "S" "S" "S" "C" "C" "S" "C" "S" "Q" "S" "S" "S" "Q" "S" "S" "S" "S" "S"
## [271] "S" "S" "S" "C" "Q" "S" "S" "S" "Q" "S" "Q" "S" "S" "S" "S" "C" "S" "S"
## [289] "S" "Q" "S" "C" "C" "S" "S" "C" "C" "S" "S" "C" "Q" "Q" "S" "Q" "S" "S"
## [307] "C" "C" "C" "C" "C" "C" "S" "S" "S" "S" "S" "S" "C" "S" "S" "Q" "S"
## [325] "S" "C" "S" "S" "S" "C" "Q" "S" "S" "S" "S" "S" "C" "S" "S" "S" "S"
## [343] "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "C" "S" "C" "S" "S" "S" "Q" "Q"
## [361] "S" "C" "C" "S" "Q" "S" "C" "C" "Q" "C" "C" "S" "S" "C" "S" "C" "S" "C"
## [379] "C" "S" "C" "C" "S" "S" "S" "S" "S" "S" "S" "Q" "C" "S" "S" "S" "C" "S"
## [397] "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "Q" "Q"
## [415] "S" "S" "S" "S" "S" "S" "C" "Q" "S" "S" "S" "S" "S" "S" "Q" "S" "S" "S"
## [433] "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "C" "S"
## [451] "S" "S" "C" "C" "S" "C" "S" "S" "S" "Q" "S" "S" "S" "S" "S" "S" "S" "S"
## [469] "Q" "C" "S" "S" "S" "C" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "C" "S"
## [487] "S" "C" "S" "S" "S" "S" "S" "C" "S" "C" "C" "S" "S" "S" "S" "Q" "Q" "S"
## [505] "S" "C" "S" "S" "S" "S" "Q" "S" "S" "C" "S" "S" "S" "Q" "S" "S" "S" "S"
## [523] "C" "C" "C" "Q" "S" "S" "S" "S" "S" "C" "C" "C" "S" "S" "S" "C" "S" "C"
## [541] "S" "S" "S" "S" "C" "S" "S" "C" "S" "S" "C" "S" "Q" "C" "S" "S" "C" "C"
## [559] "S" "S" "Q" "S" "S" "S" "S" "S" "S" "S" "C" "S" "S" "S" "S" "Q" "S" "S"
## [577] "S" "S" "C" "S" "S" "C" "S" "C" "C" "S" "S" "C" "S" "S" "S" "C" "S" "Q"
## [595] "S" "S" "S" "S" "C" "C" "S" "S" "S" "S" "C" "S" "S" "S" "C" "S" "S" "S"
## [613] "Q" "Q" "S" "S" "S" "S" "S" "S" "C" "S" "C" "S" "S" "S" "Q" "S" "S" "Q"
## [631] "S" "S" "C" "S" "S" "S" "S" "S" "S" "S" "S" "C" "S" "S" "C" "C" "S" "C"
## [649] "S" "S" "S" "S" "S" "Q" "Q" "S" "S" "Q" "S" "C" "S" "C" "S" "S" "S" "S"
## [667] "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "S" "C" "Q" "C" "S" "S"
## [685] "S" "C" "S" "S" "S" "S" "S" "C" "S" "C" "S" "S" "S" "Q" "C" "S" "C" "S"
## [703] "C" "Q" "S" "S" "S" "S" "S" "C" "C" "S" "S" "S" "S" "S" "C" "S" "Q" "S"
## [721] "S" "S" "S" "S" "S" "S" "S" "Q" "S" "S" "S" "C" "S" "S" "S" "S" "S" "C"
## [739] "S" "S" "S" "S" "C" "S" "S" "S" "S" "S" "S" "Q" "S" "S" "S" "S" "S" "S"
## [757] "S" "S" "S" "S" "S" "S" "C" "S" "S" "S" "C" "Q" "Q" "S" "S" "S" "S" "C"
## [775] "S" "S" "Q" "S" "Q" "S" "C" "S" "S" "S" "S" "S" "Q" "S" "C" "Q" "S"
## [793] "S" "C" "S" "S" "S" "S" "C" "S" "S" "S" "S" "C" "S" "S" "S" "S" "S" "S"
## [811] "S" "S" "S" "S" "S" "S" "S" "C" "S" "S" "S" "S" "S" "S" "S" "Q" "S" "C"
## [829] "Q" "S" "C" "S" "C" "S" "S" "C" "S" "S" "S" "C" "S" "S" "C" "C" "S" "S"
## [847] "S" "C" "S" "C" "S" "S" "C" "S" "S" "S" "S" "S" "C" "C" "S" "S" "S" "S"
## [865] "S" "S" "C" "S" "S" "S" "S" "S" "S" "S" "C" "C" "S" "S" "S" "C" "S" "S"
## [883] "S" "S" "S" "Q" "S" "S" "S" "C" "Q"
```

titanic_train\$Age

```
## [1] 22.00 38.00 26.00 35.00 35.00 NA 54.00 2.00 27.00 14.00 4.00 58.00
## [13] 20.00 39.00 14.00 55.00 2.00 NA 31.00 NA 35.00 34.00 15.00 28.00
## [25] 8.00 38.00 NA 19.00 NA NA 40.00 NA NA 66.00 28.00 42.00
## [37] NA 21.00 18.00 14.00 40.00 27.00 NA 3.00 19.00 NA NA NA
## [49] NA 18.00 7.00 21.00 49.00 29.00 65.00 NA 21.00 28.50 5.00 11.00
## [61] 22.00 38.00 45.00 4.00 NA NA 29.00 19.00 17.00 26.00 32.00 16.00
## [73] 21.00 26.00 32.00 25.00 NA NA 0.83 30.00 22.00 29.00 NA 28.00
## [85] 17.00 33.00 16.00 NA 23.00 24.00 29.00 20.00 46.00 26.00 59.00 NA
## [97] 71.00 23.00 34.00 34.00 28.00 NA 21.00 33.00 37.00 28.00 21.00 NA
## [109] 38.00 NA 47.00 14.50 22.00 20.00 17.00 21.00 70.50 29.00 24.00 2.00
## [121] 21.00 NA 32.50 32.50 54.00 12.00 NA 24.00 NA 45.00 33.00 20.00
## [133] 47.00 29.00 25.00 23.00 19.00 37.00 16.00 24.00 NA 22.00 24.00 19.00
## [145] 18.00 19.00 27.00 9.00 36.50 42.00 51.00 22.00 55.50 40.50 NA 51.00
## [157] 16.00 30.00 NA NA 44.00 40.00 26.00 17.00 1.00 9.00 NA 45.00
## [169] NA 28.00 61.00 4.00 1.00 21.00 56.00 18.00 NA 50.00 30.00 36.00
## [181] NA NA 9.00 1.00 4.00 NA NA 45.00 40.00 36.00 32.00 19.00
## [193] 19.00 3.00 44.00 58.00 NA 42.00 NA 24.00 28.00 NA 34.00 45.50
## [205] 18.00 2.00 32.00 26.00 16.00 40.00 24.00 35.00 22.00 30.00 NA 31.00
## [217] 27.00 42.00 32.00 30.00 16.00 27.00 51.00 NA 38.00 22.00 19.00 20.50
## [229] 18.00 NA 35.00 29.00 59.00 5.00 24.00 NA 44.00 8.00 19.00 33.00
## [241] NA NA 29.00 22.00 30.00 44.00 25.00 24.00 37.00 54.00 NA 29.00
## [253] 62.00 30.00 41.00 29.00 NA 30.00 35.00 50.00 NA 3.00 52.00 40.00
## [265] NA 36.00 16.00 25.00 58.00 35.00 NA 25.00 41.00 37.00 NA 63.00
## [277] 45.00 NA 7.00 35.00 65.00 28.00 16.00 19.00 NA 33.00 30.00 22.00
## [289] 42.00 22.00 26.00 19.00 36.00 24.00 24.00 NA 23.50 2.00 NA 50.00
## [301] NA NA 19.00 NA NA 0.92 NA 17.00 30.00 30.00 24.00 18.00
## [313] 26.00 28.00 43.00 26.00 24.00 54.00 31.00 40.00 22.00 27.00 30.00 22.00
## [325] NA 36.00 61.00 36.00 31.00 16.00 NA 45.50 38.00 16.00 NA NA
## [337] 29.00 41.00 45.00 45.00 2.00 24.00 28.00 25.00 36.00 24.00 40.00 NA
## [349] 3.00 42.00 23.00 NA 15.00 25.00 NA 28.00 22.00 38.00 NA NA
## [361] 40.00 29.00 45.00 35.00 NA 30.00 60.00 NA NA 24.00 25.00 18.00
## [373] 19.00 22.00 3.00 NA 22.00 27.00 20.00 19.00 42.00 1.00 32.00 35.00
## [385] NA 18.00 1.00 36.00 NA 17.00 36.00 21.00 28.00 23.00 24.00 22.00
## [397] 31.00 46.00 23.00 28.00 39.00 26.00 21.00 28.00 20.00 34.00 51.00 3.00
## [409] 21.00 NA NA NA 33.00 NA 44.00 NA 34.00 18.00 30.00 10.00
## [421] NA 21.00 29.00 28.00 18.00 NA 28.00 19.00 NA 32.00 28.00 NA
## [433] 42.00 17.00 50.00 14.00 21.00 24.00 64.00 31.00 45.00 20.00 25.00 28.00
## [445] NA 4.00 13.00 34.00 5.00 52.00 36.00 NA 30.00 49.00 NA 29.00
## [457] 65.00 NA 50.00 NA 48.00 34.00 47.00 48.00 NA 38.00 NA 56.00
## [469] NA 0.75 NA 38.00 33.00 23.00 22.00 NA 34.00 29.00 22.00 2.00
## [481] 9.00 NA 50.00 63.00 25.00 NA 35.00 58.00 30.00 9.00 NA 21.00
## [493] 55.00 71.00 21.00 NA 54.00 NA 25.00 24.00 17.00 21.00 NA 37.00
## [505] 16.00 18.00 33.00 NA 28.00 26.00 29.00 NA 36.00 54.00 24.00 47.00
## [517] 34.00 NA 36.00 32.00 30.00 22.00 NA 44.00 NA 40.50 50.00 NA
## [529] 39.00 23.00 2.00 NA 17.00 NA 30.00 7.00 45.00 30.00 NA 22.00
## [541] 36.00 9.00 11.00 32.00 50.00 64.00 19.00 NA 33.00 8.00 17.00 27.00
## [553] NA 22.00 22.00 62.00 48.00 NA 39.00 36.00 NA 40.00 28.00 NA
## [565] NA 24.00 19.00 29.00 NA 32.00 62.00 53.00 36.00 NA 16.00 19.00
## [577] 34.00 39.00 NA 32.00 25.00 39.00 54.00 36.00 NA 18.00 47.00 60.00
## [589] 22.00 NA 35.00 52.00 47.00 NA 37.00 36.00 NA 49.00 NA 49.00
## [601] 24.00 NA NA 44.00 35.00 36.00 30.00 27.00 22.00 40.00 39.00 NA
## [613] NA NA 35.00 24.00 34.00 26.00 4.00 26.00 27.00 42.00 20.00 21.00
```

```
## [625] 21.00 61.00 57.00 21.00 26.00 NA 80.00 51.00 32.00 NA 9.00 28.00
## [637] 32.00 31.00 41.00 NA 20.00 24.00 2.00 NA 0.75 48.00 19.00 56.00
## [649] NA 23.00 NA 18.00 21.00 NA 18.00 24.00 NA 32.00 23.00 58.00
## [661] 50.00 40.00 47.00 36.00 20.00 32.00 25.00 NA 43.00 NA 40.00 31.00
## [673] 70.00 31.00 NA 18.00 24.50 18.00 43.00 36.00 NA 27.00 20.00 14.00
## [685] 60.00 25.00 14.00 19.00 18.00 15.00 31.00 4.00 NA 25.00 60.00 52.00
## [697] 44.00 NA 49.00 42.00 18.00 35.00 18.00 25.00 26.00 39.00 45.00 42.00
## [709] 22.00 NA 24.00 NA 48.00 29.00 52.00 19.00 38.00 27.00 NA 33.00
## [721] 6.00 17.00 34.00 50.00 27.00 20.00 30.00 NA 25.00 25.00 29.00 11.00
## [733] NA 23.00 23.00 28.50 48.00 35.00 NA NA NA 36.00 21.00 24.00
## [745] 31.00 70.00 16.00 30.00 19.00 31.00 4.00 6.00 33.00 23.00 48.00 0.67
## [757] 28.00 18.00 34.00 33.00 NA 41.00 20.00 36.00 16.00 51.00 NA 30.50
## [769] NA 32.00 24.00 48.00 57.00 NA 54.00 18.00 NA 5.00 NA 43.00
## [781] 13.00 17.00 29.00 NA 25.00 25.00 18.00 8.00 1.00 46.00 NA 16.00
## [793] NA NA 25.00 39.00 49.00 31.00 30.00 30.00 34.00 31.00 11.00 0.42
## [805] 27.00 31.00 39.00 18.00 39.00 33.00 26.00 39.00 35.00 6.00 30.50 NA
## [817] 23.00 31.00 43.00 10.00 52.00 27.00 38.00 27.00 2.00 NA NA 1.00
## [829] NA 62.00 15.00 0.83 NA 23.00 18.00 39.00 21.00 NA 32.00 NA
## [841] 20.00 16.00 30.00 34.50 17.00 42.00 NA 35.00 28.00 NA 4.00 74.00
## [853] 9.00 16.00 44.00 18.00 45.00 51.00 24.00 NA 41.00 21.00 48.00 NA
## [865] 24.00 42.00 27.00 31.00 NA 4.00 26.00 47.00 33.00 47.00 28.00 15.00
## [877] 20.00 19.00 NA 56.00 25.00 33.00 22.00 28.00 25.00 39.00 27.00 19.00
## [889] NA 26.00 32.00
```

Viendo data, cantidades de NA y porcentajes de NA y “ ”

```
summary(titanic_train)
```

```
## PassengerId      Survived      Pclass         Name
## Min.   : 1.0      Min.   :0.0000   Min.   :1.000   Length:891
## 1st Qu.:223.5    1st Qu.:0.0000   1st Qu.:2.000   Class :character
## Median :446.0    Median :0.0000   Median :3.000   Mode  :character
## Mean   :446.0    Mean   :0.3838   Mean    :2.309
## 3rd Qu.:668.5    3rd Qu.:1.0000   3rd Qu.:3.000
## Max.   :891.0    Max.   :1.0000   Max.    :3.000
##
## Sex              Age              SibSp          Parch
## Length:891      Min.    : 0.42   Min.    :0.000   Min.    :0.0000
## Class :character 1st Qu.:20.12   1st Qu.:0.000   1st Qu.:0.0000
## Mode  :character Median :28.00   Median :0.000   Median :0.0000
##                  Mean   :29.70   Mean   :0.523   Mean   :0.3816
##                  3rd Qu.:38.00   3rd Qu.:1.000   3rd Qu.:0.0000
##                  Max.    :80.00   Max.    :8.000   Max.    :6.0000
##                  NA's    :177
## Ticket          Fare              Cabin          Embarked
## Length:891      Min.    : 0.00   Length:891      Length:891
## Class :character 1st Qu.: 7.91   Class :character Class :character
## Mode  :character Median :14.45   Mode  :character Mode :character
##                  Mean   :32.20
##                  3rd Qu.:31.00
##                  Max.    :512.33
##
```

```
colSums(is.na(titanic_train))
```

```
## PassengerId      Survived      Pclass         Name         Sex         Age
```

```
##      0      0      0      0      0      177
## SibSp  Parch  Ticket  Fare  Cabin  Embarked
##      0      0      0      0      0      0
```

```
colMeans(is.na(titanic_train))*100
```

```
## PassengerId  Survived  Pclass     Name     Sex      Age
##    0.00000    0.00000    0.00000    0.00000    0.00000    19.86532
## SibSp      Parch    Ticket     Fare     Cabin  Embarked
##    0.00000    0.00000    0.00000    0.00000    0.00000    0.00000
```

```
colSums(titanic_train == "", na.rm = TRUE)
```

```
## PassengerId  Survived  Pclass     Name     Sex      Age
##           0           0           0           0           0           0
## SibSp      Parch    Ticket     Fare     Cabin  Embarked
##           0           0           0           0          687           2
```

```
colMeans(titanic_train == "", na.rm = TRUE)*100
```

```
## PassengerId  Survived  Pclass     Name     Sex      Age
## 0.0000000    0.0000000    0.0000000    0.0000000    0.0000000    0.0000000
## SibSp      Parch    Ticket     Fare     Cabin  Embarked
## 0.0000000    0.0000000    0.0000000    0.0000000    77.1043771    0.2244669
```

Como se puede observar Age es la unica variable que presenta datos NA, siendo estos 177, un 19.865% de las observaciones, y Embarked presenta dos datos faltantes “”, representando el 0.224% de embarked

- Limpie los datos considerando al menos las variables: Survived, Pclass, Sex, Age, SibSp, Parch, Fare, Embarked.
- Transforme las variables categóricas en factores cuando corresponda. Selecccion de variables y transformacion de variables a factores

```
titanic <- titanic_train %>%
  dplyr::select(
    Survived,
    Pclass,
    Sex,
    Age,
    SibSp,
    Parch,
    Fare,
    Embarked
  )

titanic <- titanic %>%
  mutate(
    Survived = factor(Survived),
    Pclass = factor(Pclass),
    Sex = factor(Sex)
  )
```

- Justifique cómo manejará los valores perdidos, especialmente en Age y Embarked.

Tratamiento de valores NA

```
titanic$Age[is.na(titanic$Age)] <- median(titanic$Age, na.rm = TRUE)

table(titanic$Embarked)
```

```
##
##      C    Q    S
##  2 168  77 644

titanic$Embarked[titanic$Embarked == ""] <- "S"

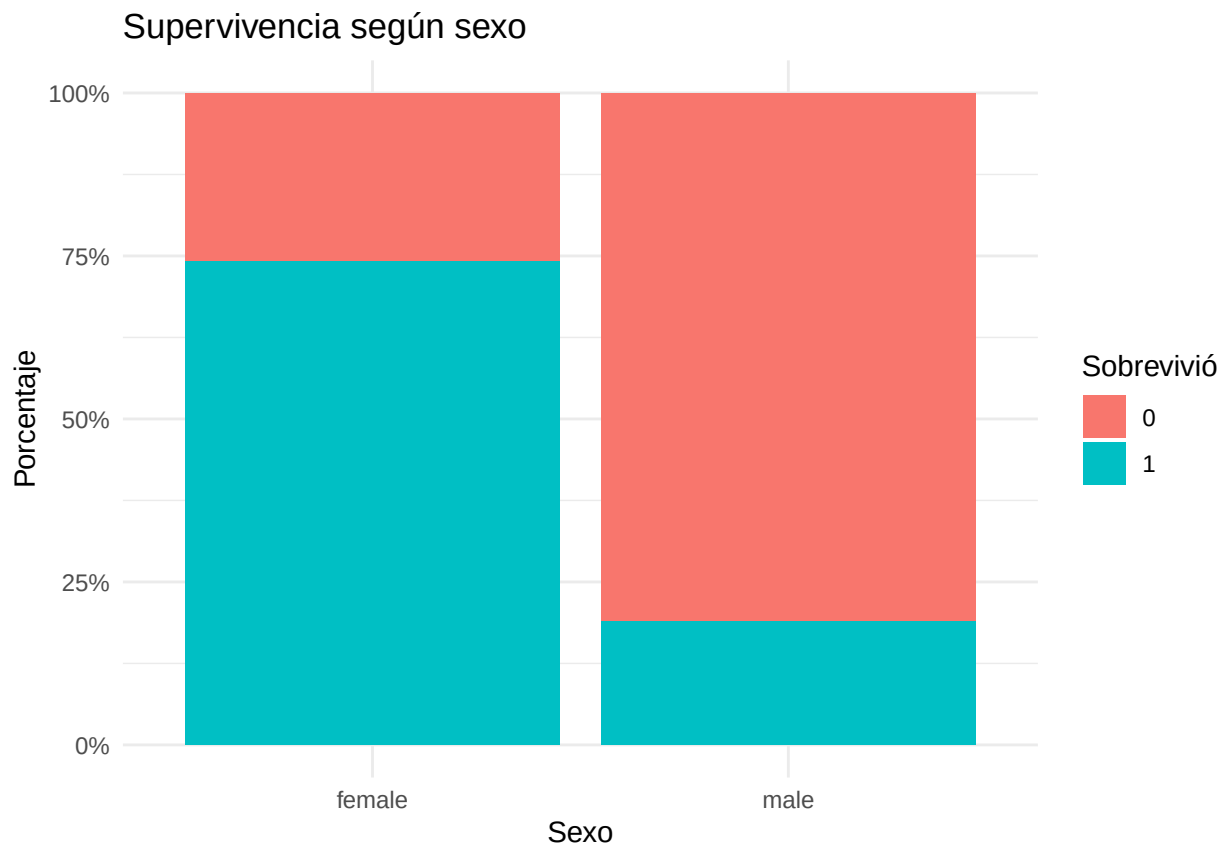
titanic$Embarked <- factor(titanic$Embarked)
```

La variable Age contiene aproximadamente un 20 % de valores faltantes. En lugar de eliminar estas observaciones, se reemplazo por la mediana de la edad para conservar el tamaño de la muestra y reducir el efecto de valores extremos. La variable Embarked presentaba únicamente dos valores perdidos (“”), por lo que se reemplazaron por la moda de la variable (S), ya que representa la opción más frecuente y minimiza el impacto sobre la distribución original de los datos.

1.2. Visualización exploratoria

Grafico 1.- Supervivencia según sexo

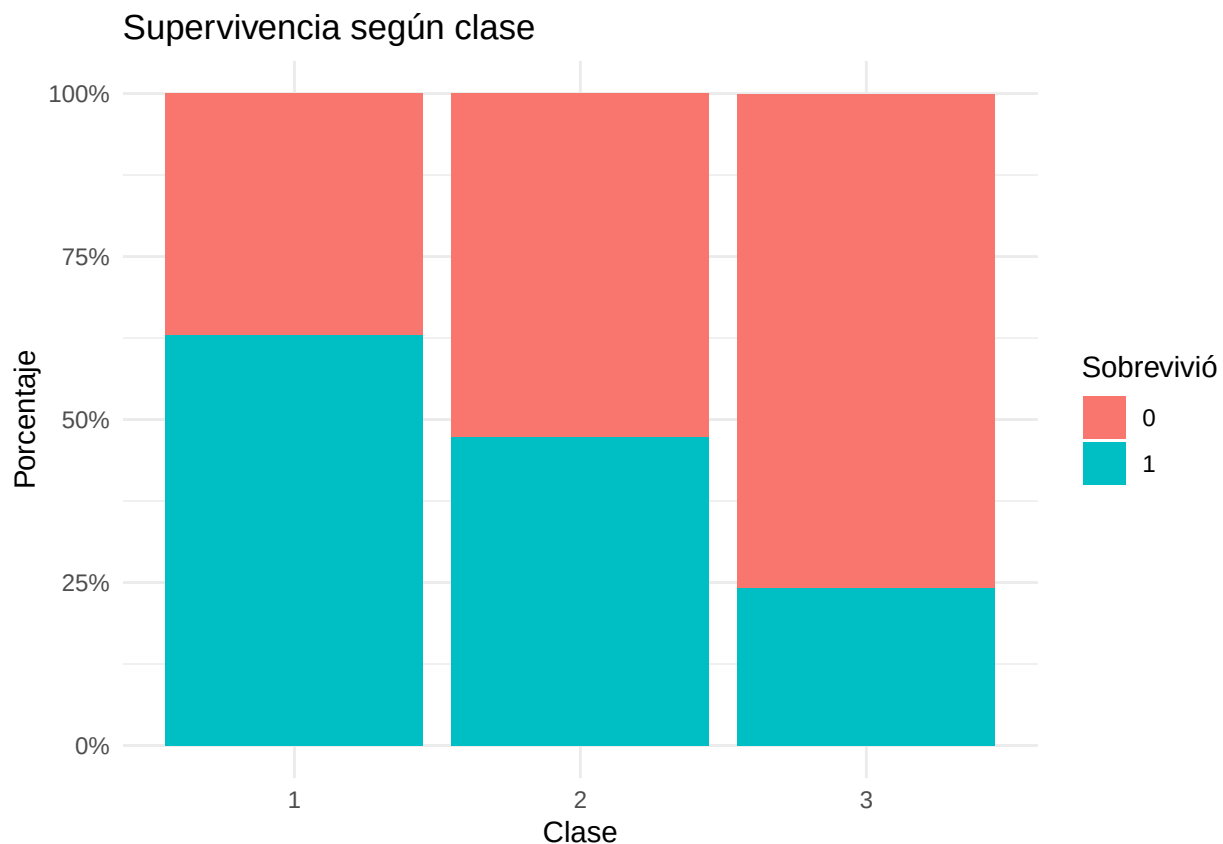
```
ggplot(titanic, aes(x = Sex, fill = Survived)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Supervivencia según sexo",
    x = "Sexo",
    y = "Porcentaje",
    fill = "Sobrevivió"
  ) +
  theme_minimal()
```



Se observa una diferencia importante entre hombres y mujeres. La proporción de mujeres que sobrevivió es considerablemente mayor que la de los hombres. Esto sugiere que el sexo podría ser uno de los predictores más importantes dentro del modelo de regresión logística.

Grafico 2.- Supervivencia según clase

```
ggplot(titanic, aes(x = Pclass, fill = Survived)) +  
  geom_bar(position = "fill") +  
  scale_y_continuous(labels = scales::percent) +  
  labs(  
    title = "Supervivencia según clase",  
    x = "Clase",  
    y = "Porcentaje",  
    fill = "Sobrevivió"  
  ) +  
  theme_minimal()
```



Los pasajeros de primera clase presentan una proporción de supervivencia considerablemente mayor que los pasajeros de tercera clase. Esto indica que la clase del boleto podría ser una variable relevante para explicar la supervivencia.

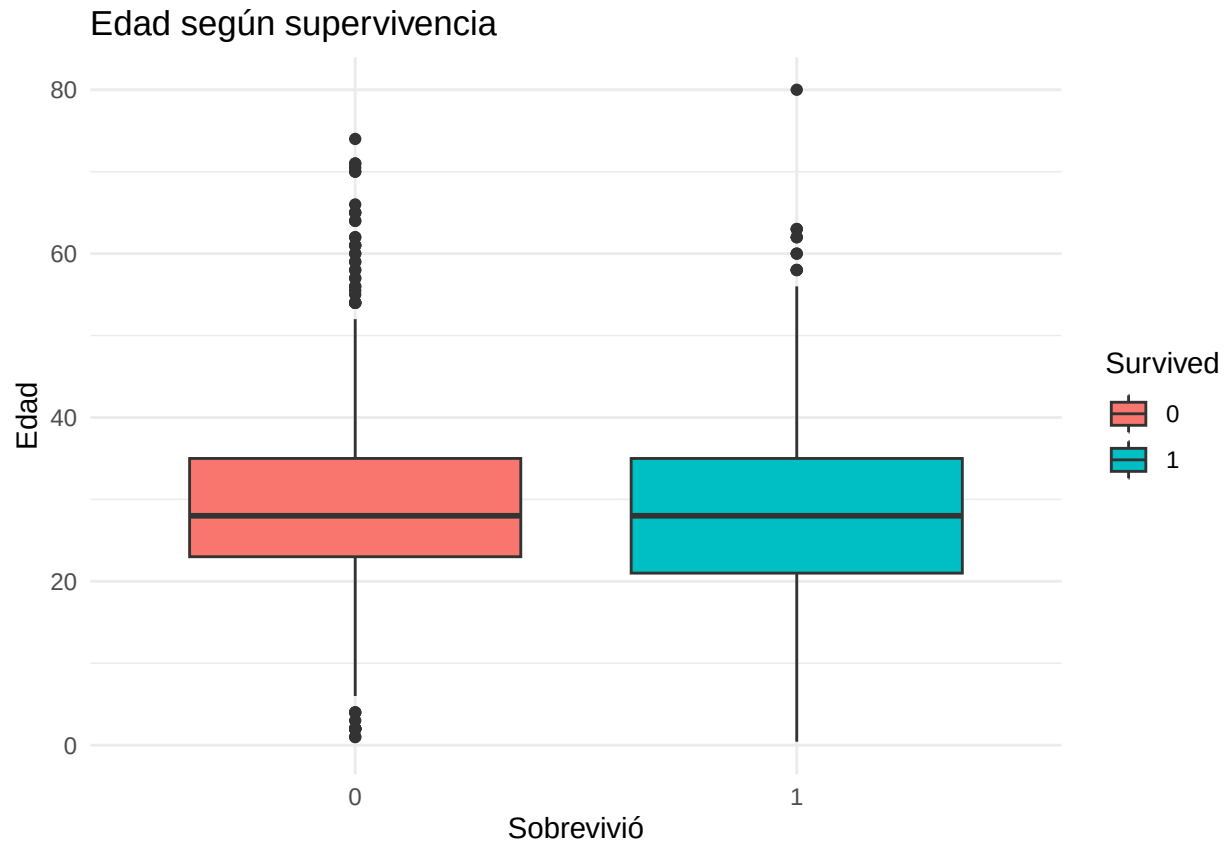
Grafico 3.- Distribución de edad según supervivencia

```
ggplot(titanic,  
  aes(x = Survived,  
    y = Age,  
    fill = Survived)) +  
  geom_boxplot() +  
  labs()
```

```

title = "Edad según supervivencia",
x = "Sobrevivió",
y = "Edad"
) +
theme_minimal()

```



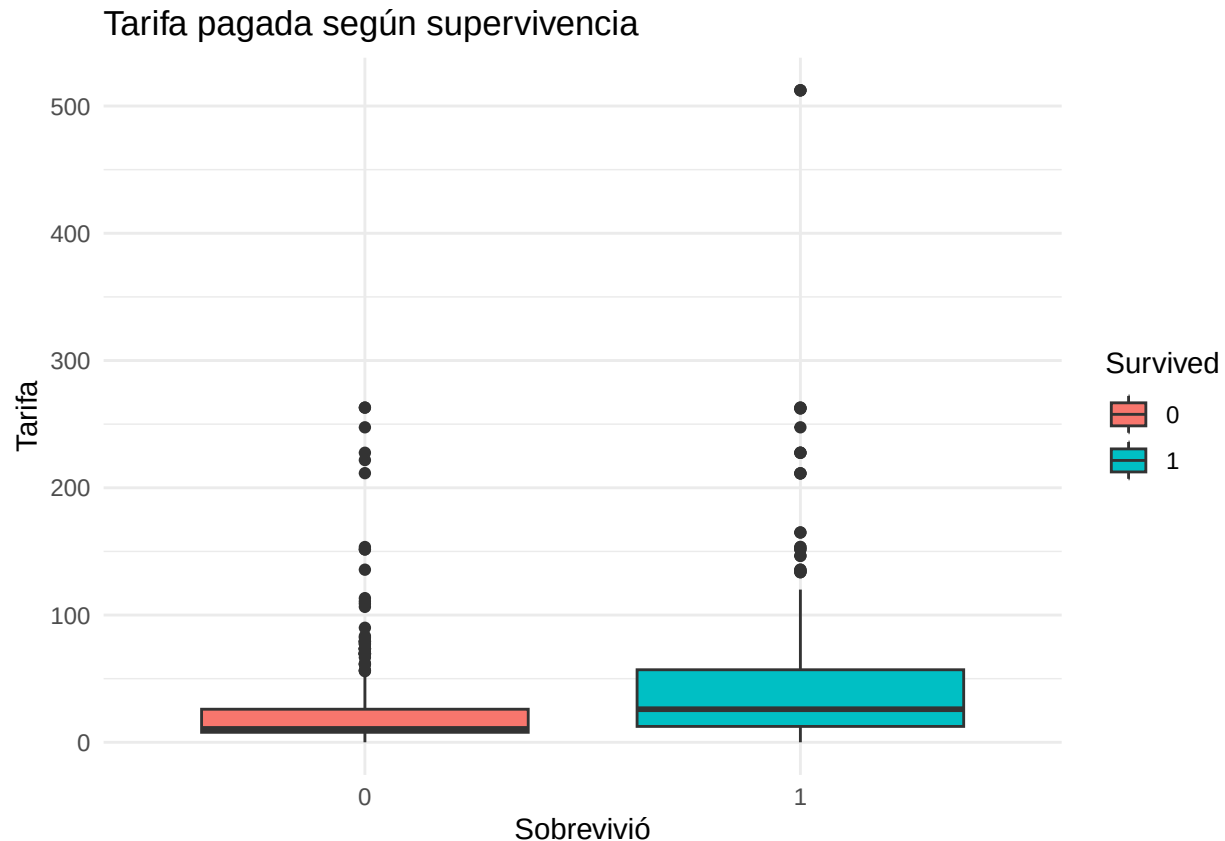
La distribución de edades presenta una fuerte superposición entre ambos grupos. Aunque los sobrevivientes parecen tener una mediana de edad ligeramente menor, la edad por sí sola no permite separar claramente a quienes sobrevivieron de quienes no.

Gráfico 4.- Tarifa pagada vs supervivencia

```

ggplot(titanic,
  aes(x = Survived,
    y = Fare,
    fill = Survived)) +
geom_boxplot() +
labs(
  title = "Tarifa pagada según supervivencia",
  x = "Sobrevivió",
  y = "Tarifa"
) +
theme_minimal()

```



Los pasajeros que sobrevivieron tienden a haber pagado tarifas más elevadas. Esto sugiere que Fare podría actuar como un indicador indirecto del nivel socioeconómico o de la clase del pasajero, variables que parecen estar relacionadas con la probabilidad de supervivencia.

El análisis exploratorio muestra que las variables Sex, Pclass y Fare presentan diferencias claras entre pasajeros que sobrevivieron y aquellos que no. En particular, las mujeres y los pasajeros de primera clase registran mayores tasas de supervivencia. La variable Age muestra diferencias menos marcadas, aunque los sobrevivientes tienden a ser ligeramente más jóvenes. Estos resultados sugieren que las variables exploradas poseen capacidad predictiva y justifican su inclusión en el modelo de regresión logística.

1.3. Modelo logístico base

```
modelo_log <- glm(
  Survived ~ Pclass + Sex + Age + SibSp + Parch + Fare + Embarked,
  data = titanic,
  family = binomial
)

summary(modelo_log)
```

```
##
## Call:
## glm(formula = Survived ~ Pclass + Sex + Age + SibSp + Parch +
##     Fare + Embarked, family = binomial, data = titanic)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.064159   0.472813   8.596  < 2e-16 ***
```

```
## Pclass2      -0.919468    0.297326   -3.092   0.00199 **
## Pclass3      -2.150048    0.297720   -7.222  5.13e-13 ***
## Sexmale      -2.719444    0.200977  -13.531   < 2e-16 ***
## Age          -0.038517    0.007855   -4.903  9.43e-07 ***
## SibSp        -0.321794    0.109193   -2.947   0.00321 **
## Parch        -0.093329    0.118856   -0.785   0.43232
## Fare          0.002339    0.002469    0.947   0.34346
## EmbarkedQ    -0.056267    0.381471   -0.148   0.88274
## EmbarkedS    -0.434226    0.239530   -1.813   0.06986 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1186.66 on 890 degrees of freedom
## Residual deviance: 785.04 on 881 degrees of freedom
## AIC: 805.04
##
## Number of Fisher Scoring iterations: 5
```

1.3.a.- ¿Qué variables son estadísticamente significativas? De acuerdo con los valores-p obtenidos en el modelo, las variables Pclass, Sex, Age y SibSp presentan evidencia estadística de asociación con la supervivencia ($p < 0.05$). Las demás variables no muestran una contribución significativa una vez consideradas conjuntamente con el resto de los predictores.

```
exp(cbind(
  OR = coef(modelo_log),
  confint(modelo_log)
))
```

```
## Waiting for profiling to be done...
##
## OR      2.5 %      97.5 %
## (Intercept) 58.21591791 23.43775342 150.03486892
## Pclass2      0.39873115 0.22199356 0.71359157
## Pclass3      0.11647860 0.06472112 0.20859270
## Sexmale      0.06591139 0.04404122 0.09692349
## Age          0.96221535 0.94723449 0.97689610
## SibSp        0.72484735 0.57887015 0.88849019
## Parch        0.91089354 0.71696424 1.14644113
## Fare         1.00234195 0.99779902 1.00776060
## EmbarkedQ    0.94528665 0.44554835 1.99180835
## EmbarkedS    0.64776558 0.40530596 1.03784575
```

1.3.b- Interprete al menos cuatro coeficientes usando odds ratios. El odds ratio de Sexmale es 0.066, lo que indica que los hombres tienen aproximadamente un 93.4 % menos de probabilidades de sobrevivir que las mujeres, manteniendo constantes las demás variables. El odds ratio de Pclass3 es 0.116, indicando que los pasajeros de tercera clase presentan aproximadamente un 88.4 % menos de probabilidades de sobrevivir que los pasajeros de primera clase. El odds ratio de Age es 0.962, por lo que cada año adicional de edad disminuye las probabilidades de supervivencia aproximadamente un 3.8 %, manteniendo constantes las demás variables. El odds ratio de Fare es 1.002, indicando que un aumento en la tarifa pagada incrementa ligeramente las probabilidades de supervivencia.

1.3.c- ¿Qué significa un odds ratio mayor que 1? ¿Y menor que 1? Un odds ratio mayor que 1 indica que la variable aumenta las probabilidades de que ocurra el evento de interés (en este caso, sobrevivir). Por el contrario, un odds ratio menor que 1 indica que la variable disminuye dichas probabilidades. Un odds ratio igual a 1 sugiere que la variable no tiene efecto sobre las probabilidades del evento.

1.3.d- ¿Qué limitaciones tiene interpretar significancia estadística sin evaluar desempeño predictivo? La significancia estadística permite determinar si existe evidencia de asociación entre una variable y la respuesta, pero no garantiza que el modelo tenga un buen desempeño predictivo. Un modelo puede contener variables altamente significativas y, aun así, clasificar mal nuevas observaciones. Por ello, además de analizar los valores p, es necesario evaluar el modelo mediante métricas como Accuracy, Sensibilidad, Especificidad, AUC ROC y la matriz de confusión utilizando un conjunto de prueba.

1.4. Selección automática por AIC

```
modelo_aic <- MASS::stepAIC(
  modelo_log,
  direction = "both",
  trace = TRUE
)
```

```
## Start:  AIC=805.04
## Survived ~ Pclass + Sex + Age + SibSp + Parch + Fare + Embarked
##
##           Df Deviance    AIC
## - Parch    1   785.67  803.67
## - Fare     1   786.01  804.01
## <none>      785.04  805.04
## - Embarked  2   789.09  805.09
## - SibSp     1   795.19  813.19
## - Age       1   810.97  828.97
## - Pclass    2   844.87  860.87
## - Sex       1  1015.02 1033.02
##
## Step:  AIC=803.67
## Survived ~ Pclass + Sex + Age + SibSp + Fare + Embarked
##
##           Df Deviance    AIC
## - Fare     1   786.37  802.37
## <none>      785.67  803.67
## - Embarked  2   789.96  803.96
## + Parch    1   785.04  805.04
## - SibSp     1   798.33  814.33
## - Age       1   811.43  827.43
## - Pclass    2   848.84  862.84
## - Sex       1  1020.34 1036.34
##
## Step:  AIC=802.37
## Survived ~ Pclass + Sex + Age + SibSp + Embarked
##
##           Df Deviance    AIC
## <none>      786.37  802.37
## - Embarked  2   791.23  803.23
## + Fare     1   785.67  803.67
## + Parch    1   786.01  804.01
## - SibSp     1   798.34  812.34
## - Age       1   812.84  826.84
## - Pclass    2   884.02  896.02
## - Sex       1  1026.84 1040.84
```

```
summary(modelo_aic)
```

```
##
## Call:
## glm(formula = Survived ~ Pclass + Sex + Age + SibSp + Embarked,
##      family = binomial, data = titanic)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.22353    0.41876  10.086 < 2e-16 ***
## Pclass2      -1.03964    0.26956  -3.857 0.000115 ***
## Pclass3      -2.30238    0.25453  -9.046 < 2e-16 ***
## Sexmale      -2.69944    0.19541 -13.814 < 2e-16 ***
## Age          -0.03874    0.00782  -4.954 7.26e-07 ***
## SibSp        -0.32720    0.10369  -3.155 0.001602 **
## EmbarkedQ     -0.05588    0.37822  -0.148 0.882536
## EmbarkedS    -0.46779    0.23610  -1.981 0.047557 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1186.66  on 890  degrees of freedom
## Residual deviance:  786.37  on 883  degrees of freedom
## AIC: 802.37
##
## Number of Fisher Scoring iterations: 5
```

```
formula(modelo_aic)
```

```
## Survived ~ Pclass + Sex + Age + SibSp + Embarked
```

1.4.a- ¿Qué variables fueron seleccionadas? Las variables seleccionadas fueron Pclass, Sex, Age, SibSp y Embarked

1.4.b- ¿El modelo seleccionado por AIC es más simple que el modelo completo? Sí, porque contiene menos predictores que el modelo completo.

```
AIC(modelo_log)
```

```
## [1] 805.0398
```

```
AIC(modelo_aic)
```

```
## [1] 802.3658
```

1.4.c- Compare el AIC del modelo completo y del modelo seleccionado. El modelo completo obtuvo un AIC de 805.04, mientras que el modelo seleccionado mediante stepAIC() obtuvo un AIC de 802.37. Dado que un menor valor de AIC indica un mejor equilibrio entre el ajuste del modelo y su complejidad, se concluye que el modelo seleccionado es preferible según este criterio. La diferencia de aproximadamente 2.67 unidades sugiere una mejora moderada al eliminar variables que aportaban poca información.

1.4.d- Discuta si un menor AIC implica necesariamente mejor capacidad predictiva. No, Un menor AIC indica un mejor equilibrio entre ajuste y complejidad dentro del conjunto de entrenamiento, pero no garantiza un mejor desempeño predictivo en datos nuevos. Para evaluar la capacidad predictiva es necesario utilizar un conjunto de prueba y métricas como Accuracy, Sensibilidad, Especificidad o AUC ROC.

1.5. Ridge y lasso

Convirtiendo las variables factores en dummies

```
model_dummy <- model.matrix(
  Survived ~ Pclass + Sex + Age + SibSp + Parch + Fare + Embarked,
  data = titanic
)[,-1]
```

Definiendo la variable de respuesta

```
resp <- as.numeric(as.character(titanic$Survived))
```

Definiendo modelo ridge

```
ridge_model <- glmnet(
  model_dummy,
  resp,
  family = "binomial",
  alpha = 0
)
```

validacion cruzada

```
cv_ridge <- cv.glmnet(
  model_dummy,
  resp,
  family = "binomial",
  alpha = 0
)
cv_ridge$lambda.min
```

```
## [1] 0.02642423
```

```
cv_ridge$lambda.1se
```

```
## [1] 0.07352698
```

Definiendo modelo lasso

```
cv_lasso <- cv.glmnet(
  model_dummy,
  resp,
  family = "binomial",
  alpha = 1
)
cv_lasso$lambda.min
```

```
## [1] 0.005309243
```

```
cv_lasso$lambda.1se
```

```
## [1] 0.02581673
```

1.5.d- Compare los valores de lambda.min y lambda.1se. lambda.min corresponde al valor de λ que minimiza el error promedio de validación cruzada. lambda.1se corresponde al mayor valor de λ cuyo error se encuentra dentro de una desviación estándar del mínimo.

En Ridge, lambda.1se (0.06699505) es mayor que lambda.min (0.02642423). En Lasso, lambda.1se (0.02581673) también es mayor que lambda.min (0.001738538).

Esto significa que lambda.1se aplica una penalización más fuerte sobre los coeficientes, produciendo modelos más simples y menos propensos al sobreajuste, aunque con una ligera pérdida en el ajuste respecto al modelo obtenido con lambda.min.

Comparando coeficientes de ridge y lasso

```
coef(
  cv_ridge,
  s = "lambda.min"
)

## 10 x 1 sparse Matrix of class "dgCMatrix"
##           lambda.min
## (Intercept)  2.68548102
## Pclass2     -0.33441805
## Pclass3     -1.36854050
## Sexmale     -2.19711890
## Age         -0.02455830
## SibSp       -0.22146549
## Parch       -0.06416672
## Fare        0.00433992
## EmbarkedQ   -0.07545778
## EmbarkedS   -0.42264277

coef(
  cv_lasso,
  s = "lambda.min"
)

## 10 x 1 sparse Matrix of class "dgCMatrix"
##           lambda.min
## (Intercept)  3.462226223
## Pclass2     -0.667731813
## Pclass3     -1.882083384
## Sexmale     -2.553987681
## Age         -0.030983084
## SibSp       -0.258114501
## Parch       -0.046381905
## Fare        0.002250285
## EmbarkedQ   .
## EmbarkedS   -0.381867192
```

1.5.e- Lasso eliminó la variable EmbarkedQ, indicando que esta aporta poca información para explicar la supervivencia cuando se consideran conjuntamente las demás variables: Pclass, sex, age, sibsp, parch, fare, embarkedS y embarkedC.